

MUSTHOFA JOKO ANGGORO

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RESEARCH STATEMENT

Aspiring PhD researcher in edge AI systems with hands-on experience in hardware-software co-design (FPGA/VHDL), distributed systems architecture, and resource-constrained computing. Seeking to advance system-level optimizations for AI at the edge with sustainability objectives, bridging computer architecture and distributed systems to enable efficient edge-fog-cloud continuum deployments.

RESEARCH INTERESTS

Computer systems architecture, distributed systems, operating systems, hardware-software co-design — with particular interest in edge AI systems, resource-constrained computing, and energy-efficient system design across the computing continuum

EDUCATION

University of Indonesia (UI)

Bachelor of Computer Science, Information Systems

Jakarta, Indonesia

Expected Graduation: Aug 2026

• **GPA: 3.84/4.00**

- **Thesis Advisor:** Prof. Petrus Mursanto (Full Professor, former Dean of Faculty of Computer Science)
- **Thesis:** Implementation of Morris Mano Computer Architecture with Enhanced Jump Branching Capabilities on Xilinx Spartan-3AN FPGA Platform (XC3S700AN) using VHDL
 - Extended baseline architecture by implementing jump branching in datapath for complex control flow operations
 - Designed and optimized digital logic circuits for hardware-level branching mechanisms
 - Investigated hardware resource utilization and timing constraints for branching logic optimization
 - Synthesized and tested on physical FPGA hardware (not just simulation), validating functional correctness and performance
 - Mainly based on M. Morris Mano's Computer System Architecture
- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

Nanyang Technological University (NTU) & ITS

Singapore & Surabaya

INSPIRASI-NTU Summer Program 2025 (LPDP Scholarship Awardee)

July 2025

- Conducted research on smart port automation, IoT sensor networks, and energy-efficient deployments in distributed maritime systems
- Produced technical report proposing tiered port automation framework with renewable energy utilization and carbon emission reduction

EXPERIENCE

Software Engineer Intern

Traveloka (Travel Technology Company)

Systems Engineering & Backend Development

Aug 2025 - Present

- Resource optimization, distributed system architecture, fault tolerance, cloud infrastructure (AWS, Terraform), containerization (Docker, Kubernetes)

SKILLS

Systems Programming: C, Python, Java, VHDL (hardware description), Go

Hardware & Architecture: FPGA development (Xilinx Spartan-3AN XC3S700AN), VHDL, digital logic design, computer architecture implementation, hardware-software co-design

Machine Learning: Scikit-learn, TensorFlow, supervised/unsupervised learning, model evaluation, imbalanced learning

Distributed Systems: Microservices architecture, containerization (Docker, Kubernetes), cloud infrastructure (AWS, Terraform)

Operating Systems: Linux, Unix — OS internals, process management, memory management, system-level programming

RESEARCH & TECHNICAL PROJECTS

Used Car Sales Prediction System

Course Project - AI & Data Science

Systems & ML Integration

Apr 2025 - May 2025

- Designed end-to-end predictive analytics system integrating multiple ML algorithms (Gradient Boosting, Random Forest, Logistic Regression, K-Means clustering)
- Implemented data preprocessing and feature engineering workflows to handle imbalanced datasets
- Applied performance evaluation methodologies and statistical analysis for model optimization